

PES University - Department of Computer Science & Engineering

Lab 1: Requirements Engineering & UML Use-Case Modelling

Combined Deliverables Specification

Problem Statement #25:	Municipal Solid Waste Route Optimizer
Domain:	Smart Cities, Transport & Logistics
Target Stakeholders / Actors:	Sanitation Supervisor, Truck Driver, Citizen, IoT Sensor System

1. Complete Requirements Specification Table

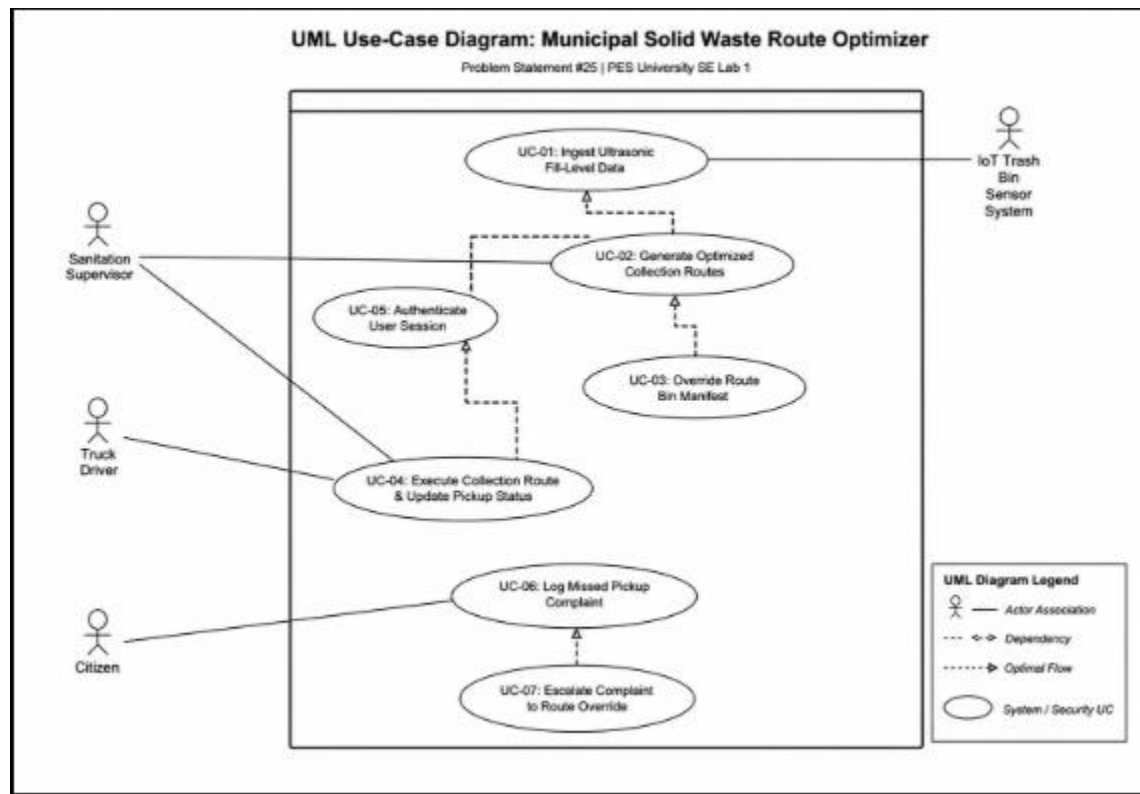
The table below details exactly 5 Functional Requirements (FR-001 to FR-005) and 2 Non-Functional Requirements (NFR-001 & NFR-002) for the Municipal Solid Waste Route Optimizer system.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall generate optimized daily collection routes including only bins that have exceeded an 80% fill threshold.	High	Pass: Route excludes bins < 80% full unless manually tagged; Fail: Overflowing bin omitted from driver manifest.	Core route planning functionality to minimize truck fuel consumption and operational overhead.
FR-002	Functional	The system shall ingest ultrasonic sensor fill-level data transmitted from smart trash bins every 15 minutes and update bin status in real time.	High	Pass: Telemetry payload updates bin fill level indicator within 3 seconds of receipt; Fail: Sensor data delayed over 1 minute or dropped without retry log.	Ensures accurate and up-to-date bin fill data for route calculation algorithms.
FR-003	Functional	The system shall allow citizens to log missed-pickup complaints with photo evidence and geolocation details.	Medium	Pass: Complaint ticket created with photo, timestamp, and location tag, returning a tracking ID to user; Fail: Complaint logged without required geolocation tag.	Provides citizen accountability and alerts supervisors to overlooked or blocked bins.
FR-004	Functional	The system shall enable the Sanitation Supervisor to manually tag/override bin collection status and reassign routes to drivers.	Medium	Pass: Supervisor-tagged bin (< 80% full) is successfully added to active truck manifest and highlighted; Fail: Manual tag overridden or rejected by automated optimizer.	Gives supervisors operational control during emergencies, special events, or complaint escalation.
FR-005	Functional	The system shall provide Truck Drivers with turn-by-turn navigation updates and allow them to mark bins as 'Collected' or 'Blocked/Inaccessible'.	High	Pass: Tapping 'Collected' updates bin status to empty (< 5%) in central portal immediately; Fail: Status fails to sync to central portal when driver is online.	Closes operational feedback loop between drivers on route and central management portal.
NFR-001	Non-Functional (Performance & Security)	Route recalculation for a fleet of 50 trucks across 10,000 bins shall execute in under 45 seconds.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load; Fail: Recalculation exceeds 45 seconds	Prevents dispatch bottlenecks during morning shifts and dynamic mid-day route

				or causes service timeout.	updates.
NFR-002	Non-Functional (Availability & Reliability)	The portal telemetry ingestion engine shall maintain 99.9% uptime and operate gracefully in offline mode for truck driver devices.	High	Pass: Telemetry service uptime >= 99.9% monthly; driver app stores up to 100 offline pickup updates and syncs automatically upon reconnection; Fail: Offline updates lost during network disconnect.	Garbage collection occurs in areas with varying cellular connectivity; data loss must be prevented.

2. UML Use-Case Diagram

The diagram below models all system actors, primary use cases, system boundary, and stereotypes («include» and «extend» relationships). An editable version is saved as Lab1_Use_Case_Diagram.drawio.



UML Stereotypes & Associations Breakdown

• «include» Relationships:

- Generate Optimized Collection Routes (UC-02) «include» Authenticate User Session (UC-05)
- Execute Collection Route (UC-04) «include» Authenticate User Session (UC-05)
- Generate Optimized Collection Routes (UC-02) «include» Ingest Ultrasonic Fill-Level Data (UC-01)

• «extend» Relationships:

- Override Route Bin Manifest (UC-03) «extend» Generate Optimized Collection Routes (UC-02)
- Escalate Complaint to Route Override (UC-07) «extend» Log Missed-Pickup Complaint (UC-06)

3. Core Use-Case Flow Specification (UC-02)

Use Case Name:	Generate Optimized Collection Routes (UC-02)
Primary Actor:	Sanitation Supervisor
Secondary Actor(s):	IoT Trash Bin Sensor System, Truck Driver (Recipient)
Short Description:	The Sanitation Supervisor initiates automated route planning. The system ingests bin fill telemetry, identifies bins $\geq 80\%$ full (or tagged), executes route optimization solver, and publishes manifests to drivers.

Preconditions & Postconditions

Preconditions:

1. Sanitation Supervisor is authenticated and logged into the portal dashboard.
2. Ultrasonic bin fill-level telemetry has been ingested into the system database within the last 15 minutes.
3. Active garbage trucks, driver schedules, and depot locations are configured in the system.

Postconditions:

1. Optimized collection route manifests are generated and published to driver mobile apps.
2. Targeted trash bins (fill level $\geq 80\%$ or supervisor-tagged) are locked into the active collection schedule.
3. System logs route dispatch metrics (total distance, estimated travel time, expected fuel consumption).

Main Success Scenario

- Step 1:** Sanitation Supervisor selects 'Generate Daily Routes' from the portal dashboard.
- Step 2:** System retrieves current ultrasonic fill-level telemetry for all city trash bins from the IoT database.
- Step 3:** System filters and identifies target bins meeting or exceeding the 80% fill threshold, plus any supervisor manually tagged bins.
- Step 4:** System executes the Vehicle Routing Problem (VRP) optimization algorithm, factoring in truck capacities, depot starting points, traffic parameters, and bin GPS coordinates.
- Step 5:** System presents draft collection routes on an interactive map along with route metrics (total distance, estimated completion time, driver assignments).
- Step 6:** Sanitation Supervisor reviews the draft routes and approves dispatch.
- Step 7:** System generates digital manifests and transmits turn-by-turn navigation schedules to assigned drivers' mobile devices.
- Step 8:** System updates bin statuses in the portal database to 'Scheduled for Pickup'.
- Step 9:** System records route dispatch timestamp and displays confirmation notification to the supervisor.

Step 10: Use case terminates successfully.

Alternate Flows

Alternate Flow 4a: Primary Optimization Solver Timeout / Recalculation Fallback (NFR-001)

4a1. System detects route optimization solver execution time exceeding the 45-second performance threshold (NFR-001).

4a2. System logs solver warning and automatically triggers fallback to cluster-based heuristic routing algorithm.

4a3. Heuristic route is computed within 10 seconds and rendered with a warning notification: 'Sub-optimal route generated due to solver timeout.'

4a4. Sanitation Supervisor reviews heuristic route and confirms dispatch or manually adjusts bin allocations.

4a5. Flow resumes at Step 7 of Main Success Scenario.

Alternate Flow 3a: No Bins Exceed Fill Threshold

3a1. System query identifies 0 trash bins meeting the 80% fill threshold and 0 manually tagged bins.

3a2. System displays informational alert: 'No collection required today based on current threshold (80%).'

3a3. Supervisor acknowledges alert or adjusts fill threshold parameter to 70%.

3a4. Flow resumes at Step 3 of Main Success Scenario with updated parameters.